

Practical no. 22

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Class :- CO3K – B

Subject :- DSU

Program :-

```
#include <stdio.h>
#include <conio.h>
#include <stdlib.h>
struct Node {
    int data;
    struct Node* next;
};
struct Node* front = NULL;
struct Node* rear = NULL;
void insert(int value) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
    if (newNode == NULL) {
        printf("\nMemory not available!\n");
        return;
    }
    newNode->data = value;
    newNode->next = NULL;
    if (rear == NULL) {
        front = rear = newNode;
    }
    else {
        rear->next = newNode;
        rear = newNode;
    }
    printf("\n%d inserted into queue.\n", value);
}
void delete() {
    struct Node* temp = front;
    if (front == NULL) {
        printf("\nQueue is Empty!\n");
        return;
    }
    printf("\n%d deleted from queue.\n", front->data);
    front = front->next;
    if (front == NULL) {
```

```

rear = NULL;
}
free(temp);
}
void display() {
    struct Node* temp = front;
    if (front == NULL) {
        printf("\nQueue is Empty!\n");
        return;
    }
    printf("\nQueue elements: ");
    while (temp != NULL) {
        printf(" %d ", temp->data);
        temp = temp->next;
    }
    printf("\n");
}
void main() {
    int choice, value;
    while (1) {
        printf("\n--- Queue Menu (Linked List) ---\n");
        printf("1. Insert (Enqueue)\n");
        printf("2. Delete (Dequeue)\n");
        printf("3. Display\n");
        printf("4. Exit\n");
        printf("\nEnter choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1:
                printf("\nEnter value to insert: ");
                scanf("%d", &value);
                insert(value);
                break;
            case 2:
                delete();
                break;
            case 3:
                display();
                break;
            case 4:
                exit(0);

```

```
default:
printf("\nInvalid choice!\n");
}
}
getch();
}
```

OUTPUT :-

```
--- Queue Menu (Linked List) ---
1. Insert (Enqueue)
2. Delete (Dequeue)
3. Display
4. Exit

Enter choice: 1

Enter value to insert: 10

10 inserted into queue.
```

```
--- Queue Menu (Linked List) ---
1. Insert (Enqueue)
2. Delete (Dequeue)
3. Display
4. Exit

Enter choice: 3

Queue elements: 10
```

```
--- Queue Menu (Linked List) ---
1. Insert (Enqueue)
2. Delete (Dequeue)
3. Display
4. Exit

Enter choice: 2

10 deleted from queue.
```